Progressive Context-Aware Dynamic Network for Salient Object Detection in Optical Remote Sensing Images

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***Abstract*— Although remarkable progress has been made in salient object detection (SOD) in optical remote sensing images (RSIs), the static network design paradigm adopted by existing methods would limit their adaptability to large variations in remote sensing scenes as well as object appearances. In contrast, we explore this research issue from the perspective of generating dynamic network filters, in which the parameters are conditioned on specific scene- and location-level contexts. In this article, we propose a progressive context-aware dynamic network (PCD-Net) for SOD in RSIs, which adaptively captures context information and adjusts its filtering parameters for saliency detection. PCD-Net adopts an encoder–decoder architecture, in which encoded feature representations are progressively decoded by a newly proposed dynamic module, namely, pyramid scene- and location-sensitive dynamic (PSLD) filtering module, to generate saliency representations. Furthermore, to transfer effective features from the encoder to the decoder, we construct a dynamic transfer attention (DTA) module to control the interference between the encoder and the decoder in a more flexible way. Extensive evaluations of two commonly used benchmarks demonstrate the superiority of the proposed method against the existing state-of-the-art methods.**

***Index Terms*— Dynamic filtering networks, filtration mechanism, remote sensing images (RSIs), salient object detection (SOD).**

I. Introduction

SALIENT object detection (SOD) aims at extracting the most attention-grabbing objects in an image, which serves as an important preprocessing tool in various vision tasks, such as image and video compression [1], image retargeting [2], image quality assessment [3], [4], object recognition [5], and so on. SOD in natural scene images (NSIs) has been thoroughly studied over the last decade and numerous methods have been proposed [6], [7], [8], [9], [10], [11]. In contrast, SOD in optical remote sensing images (RSIs) has emerged as a new topic and has started receiving attention from the research community recently. SOD in RSIs has increasingly practical values, especially for applications in remote sensing scenarios, such as scene classification [12], object localization [13], anomaly discovery, geographical navigation, and so on. Therefore, it is necessary to investigate RSI-SOD academically.

SOD in NSIs (NSI-SOD) has witnessed great development with the application of convolutional neural networks (CNNs) in recent years [10], [11], [14]. It is natural to extend cutting-edge NSI-SOD methods to RSI-SOD. However, the characteristics of optical RSIs are highly different from that of NSIs. While NSIs are captured by cameras on the ground, optical RSIs are collected by remote imaging sensors mounted on airplanes or satellites. The former is usually conducted by human photographers, whereas the latter is with minimal human intervention. Therefore, RSI-SOD presents specific challenges due to the inherent characteristics of optical RSIs, e.g., varying scales of objects, arbitrary orientations, complex topology structures, unstable imaging conditions, etc. Directly applying NSI-SOD methods to RSI-SOD would lead to unsatisfactory performance. As shown in the third column of Fig. 1(a), a competitive NSI-SOD method, EGNet [15], fails to accurately extract salient objects with large scale-variations and with complex topologies in RSIs.

To address the above-mentioned challenges, a number of specialized methods for RSI-SOD have been proposed recently, which can be categorized into hand-designed feature-based methods and deep neural network-based methods. The former utilizes contrast [16], sparsity representation [17], statistical color information [18], [19], and multiple feature descriptors fusion [20] to measure saliency in RSIs. The latter focuses on constructing end-to-end learning models by employing CNNs, which have gained huge performance improvement compared with the hand-designed methods. Most CNN-based methods [21], [22], [23], [24] concentrate on designing multiscale feature fusion strategies to tackle the scale variation issue. LVNet [21], EMFI-Net [25] conduct feature extraction from multiresolution inputs and fuse them in a top-down manner. PDFNet [23], SARNet [26], CorrNet [27] thoroughly exploit the complementarities between multilevel features extracted from CNNs for RSI-SOD. There also have been attempts to utilize edge cues to boost the performance of SOD, such as MJRBM [28] and ERPNet [29]. In addition, some works [22], [30], [31] leverage on attention mechanism

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to model context information and eliminate background clutter. For example, DAFNet [22] propagates an attention mechanism from shallow layers to deep layers, exploiting long-range relations in context modeling, and RRNet [30] constructs a parallel multiscale attention scheme to recover the detailed information of salient objects.

Despite remarkable progress in RSI-SOD has been made, there has been an insufficient investigation into the adaptability of models to handle diversified variations presented in RSIs, which may limit further improvement. First, existing methods adopt the stationary network paradigm, which cannot adjust its parameters to adapt to specific samples during inference. Second, although the encoder-decoder architecture has been widely adopted, there is a lack of filtration mechanisms between them for transferring effective features while suppressing background interference. Due to these reasons, the trained models can not well adapt to RSIs, where salient objects usually appear in varying scales and irregular topologies [as shown in Fig. 1(a)]. In this work we explore RSI-SOD from the perspective of dynamic filtering networks inspired by [32]. It is challenging and nontrivial to apply dynamic filter networks to RSI-SOD since salient objects in RSIs present large variations in diversified aspects, including types, scales, locations, and topologies. Meanwhile, current dynamic networks designed for semantic segmentation and object detection [33], [34] fail to consider scene-correlated context information as well as multiscale context information in dynamic filter generation. The schematic comparison of our work with existing RSI-SOD methods is shown in Fig. 1(b). Our work is distinguished by its adaptability to various scenes and objects in RSIs, which is acquired by leveraging the contextual information in both scene-level and location-level representations to generate filer parameters dynamically.

In this article, we propose a progressive context-aware dynamic network (PCD-Net) for RSI-SOD, which adaptively captures context information and adjusts its filtering parameters accordingly. The proposed network architecture adopts an encoder-decoder architecture, where the encoder is borrowed from well-known feature extraction networks, e.g., VGG [35], and the decoder progressively recovers saliency representations in a sample-specific manner. Our PCD-Net has two novel modules: First, to transfer effective features from the encoder to the decoder, we propose a dynamic transfer attention (DTA) module, which suppresses irrelevant features and propagates informative features. DTA calculates an attention map simultaneously from local spatial context, global spatial context, and channel context, and then uses it to filtrate the encoder features. Second, we propose a pyramid scene- and location-sensitive dynamic (PSLD) filtering module to progressively generate saliency results layer-by-layer. PSLD generates filtering parameters dynamically conditioned on the filtrated features. Specifically, it adopts a pyramid structure to enable flexible context modeling and utilizes both scene-level and location-level context for filter generation.

Our work has the following contributions:

1) We explore SOD in RSIs from the perspective of dynamic filtering networks, and propose a PCD-Net which adaptively captures context information and adjusts its filtering parameters conditioned on specific inputs, showing strong adaptability to various remote sensing scenarios.
2) We propose a PSLD filtering module to generate filters dynamically and apply them to saliency calculation. Both scene-level and location-level contexts are incorporated in the filter generation process; meanwhile, the output saliency captures rich multiscale contexts with the pyramid structure.
3) A feature filtration mechanism, which is realized by a proposed DTA module, is constructed to control the interference between the encoder and the decoder.

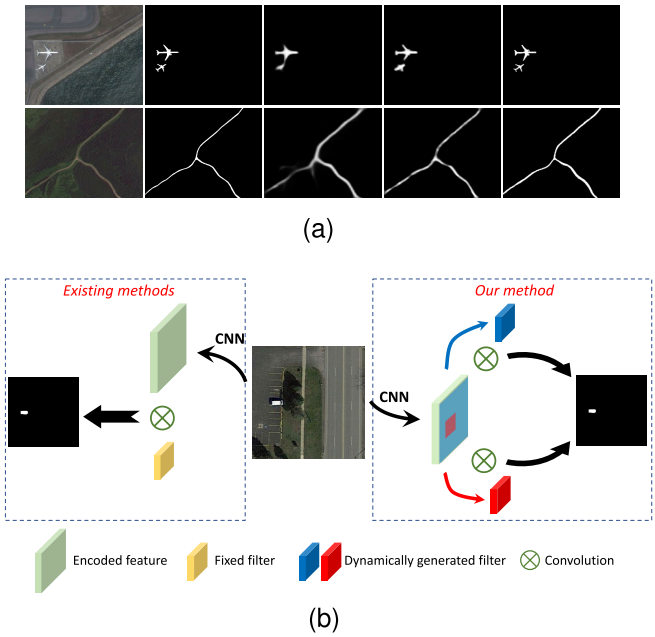


Fig. 1. (a) Visualization of saliency maps generated by different methods. From left to right column: RSIs, ground truth, and saliency maps generated by EGNet [15], DAFNet [22], and Our method. (b) Overall pipeline comparison of our proposed method (right-hand side part) with the existing methods (left-hand side part). Different colors overlaying on the CNN feature maps (right-hand side part) correspond to the contextual information of different scales on which the dynamic kernels are conditioned. Blue: scene-level context, Red: location-level context.

The rest of this article is organized as follows. The related works on SOD are reviewed in Section II. Our proposed methodology is elaborated in Section III. In Section IV, experimental comparison and ablation study are presented. Finally, the conclusion is drawn in Section V.

II. Related Work

A. Salient Object Detection in Natural Scene Images

NSI-SOD has witnessed great development in the last decade. Early methods focus on designing hand-crafted features to measure saliency from the perspective of contrast [6], [7], [36], object prior [37], background prior [8], [38], surroundedness [9]. Some representative works are reviewed in the following. Achanta et al. [6] define salient region detection as producing saliency maps with well-defined boundaries of salient objects, and propose a simple global contrast descriptor to derive saliency map. Cheng et al. [7] propose a region contrast-based method to detect salient

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regions in images, and further utilize salient regions to perform object segmentation. Perazzi et al. [36] employ uniqueness and spatial distribution to measure contrast, and formulate saliency estimation using high-dimensional Gaussian filters. Jiang et al. [39] integrate regional contrast, regional property, and regional backgroundness descriptors together to form a saliency map. Zhang and Sclaroff [9] propose to produce a saliency map by segmenting an image into a set of binary images and analyzing the topological structure.

As CNNs have shown their power in image recognition, many researchers have deployed CNNs to NSI-SOD. Hou et al. [10] construct densely connected shortcuts on the network side outputs and integrate multiscale features to form saliency results. Zhao et al. [11] propose a dual-gated mechanism to control the interference between the encoder network and the decoder network, thus suppressing irrelevant features and enhancing discriminative features. Liu et al. [14] exploit the potential of pooling in SOD and integrate two pooling-based modules into a feature pyramid network (FPN) for saliency detection. Zhao et al. [15] utilize salient edge as an auxiliary cue to highlight the salient objects with clear boundaries. Wu et al. [40] propose a bidirectional cross-reference network that facilitates cooperative learning between SOD and salient edge detection. Li et al. [41] deploy attention mechanisms in a densely connected architecture to propagate contextual information selectively among multiple network levels.

These methods have achieved excellent performance for NSI-SOD. However, applying them directly to RSIs would obtain suboptimal results, as there is a large domain gap between NSIs and RSIs. It is desirable to design specialized models for the RSI-SOD task.

B. Salient Object Detection in Remote Sensing Images

SOD in RSIs (RSI-OSD) has emerged as a challenging task to detect salient regions in RSIs. Early methods focus on utilizing contrast, sparsity representation, statistical color information, and so on, to measure saliency in RSIs. Huang et al. [16] propose a contrast-based dictionary learning method to construct a saliency dictionary, which is leveraged to calculate saliency maps. Zhao et al. [17] construct two dictionaries to characterize the global and background information and use sparsity representation to obtain saliency prediction. Zhang et al. [18], [19] propose to measure saliency in RSIs based on the statistical color histogram. They also utilize multiple feature descriptors, such as color, intensity, texture, and so on, and incorporate them into saliency representation by a self-adaptive fusion model [20].

Recently, CNNs have been applied to RSI-SOD and have witnessed excellent performance improvement. Li et al. [21] construct a diversified dataset for RSI-SOD, opening the gate for CNN-based RSI-SOD. In addition, they propose a nested architecture for perceiving salient objects from multiresolution inputs. Cong et al. [30] propose a relational reasoning network to infer the semantic relationship for salient object localization and use a paralleled multiscale attention module to restore detailed object information. Zhang et al. [22] propose to propagate global attention from shallower layers to deeper layers, exploiting the potential of the long-range relations in feature learning. Li et al. [23] connect multiscale pathways in both top-down and down-top manners to facilitate cross-scale feature communication. Some works are utilizing auxiliary edge cues to better highlight the whole salient object regions. Tu et al. [28] propose a joint object and boundary learning model for simultaneously optimizing multiscale salient region and boundary features. Zhou et al. [25] propose to perform multiscale feature integration under the guidance of salient edge information. Some works aim at designing lightweight RSI-SOD models that are friendly to real-world applications. Shen et al. [24] propose a highly squeezed module and integrate it into an encoder–decoder architecture. Li et al. [27] propose a coarse-to-fine strategy that uses an initial saliency map computed by high-level semantic correlation to guide low-level feature refinement. Huang et al. [26] propose to roughly locate salient objects by aggregating multiple high-level features and use semantic information to guide the integration of subsequent features in a step-by-step manner.

Current specialized methods for RSI-SOD are dedicated to designing multiscale feature extraction and fusion strategies to tackle the issues of scale variation, complex topology, and so on, by constructing static network architectures in which filter parameters are fixed during inference. Such a design paradigm would limit their adaptability to large variations presented in remote sensing scenes as well as object appearances. In contrast, we explore RSI-SOD from the perspective of dynamic filtering networks and propose a novel model that can leverage the rich context of specific inputs to adjust its filtering parameters.

C. Dynamic Filtering Networks

The conception of dynamic filtering is first introduced in the task of video and stereo prediction [32]. Dynamic filters that are conditioned on input are used for feature extraction and enhancement. Recently, it has been deployed in dense-pixel prediction tasks, e.g., semantic segmentation and object detection. He et al. [33] construct a multiscale dynamic convolutional head for semantic segmentation, which generates context-aware filters by global pooling. Pang et al. [34] propose a dynamic dilated pyramid module (DDPM) to generate filter kernels with different dilation rates and then apply it for multimodal feature fusion in RGB-D SOD task. Zhang et al. [42] propose to use dynamic filtering in video SOD and generate dynamic kernels by adjacent frame features. Nirkin et al. [43] propose a novel dynamic patchwise convolution for semantic segmentation, which generates weights that vary per input and spatial location. Zhang et al. [44] represent the local patch labels using neural networks and make use of dynamic convolutions to parameterize these neural networks for semantic segmentation. Our work is distinguished from the existing dynamic filtering networks in two aspects: first, we consider multirange contexts, i.e., scene- and location-level contexts, in dynamic filter generation process, which are responsible for capturing scene- and location-level variations in RSIs, respectively; second,

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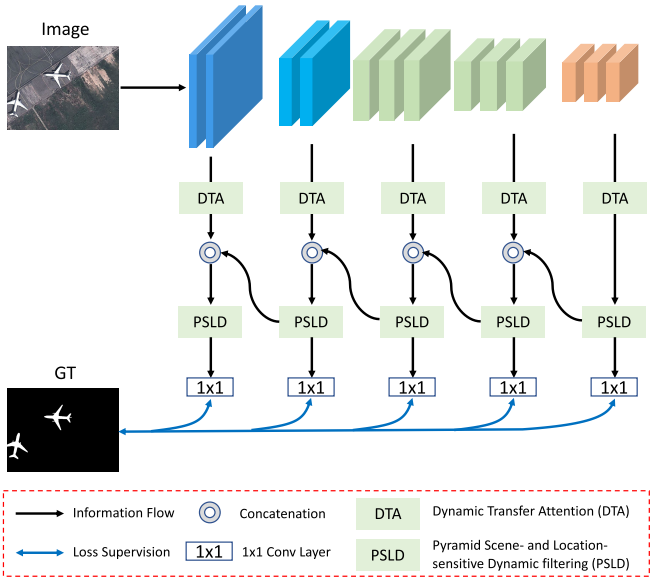


Fig. 2. Overall architecture of our proposed method.

we utilize a filtration mechanism to adaptively control the interference between the encoder and the decoder parts, facilitating transferring effective features to the decoder. The detailed pipeline comparison of our method against existing dynamic filtering networks can be found in Section III-E.

III. Proposed Method

A. Architecture Overview

The overall architecture of our proposed method is presented in Fig. 2. It follows the encoder-decoder architecture [45], [46], [47] that is widely adopted in dense pixel prediction tasks such as semantic segmentation and object detection. Most importantly, our proposed method has two distinctive merits: first, there exists a feature filtration mechanism between the encoder and the decoder (instantiated as a DTA module), which aims to suppress redundant useless features and transfer saliency features selectively; second, the feature decoding process is conducted progressively from top-level to bottom-level by a pyramid scene and location-sensitive dynamic PSLD filtering module, which generates filtering parameters dynamically conditioned on the input features. Specifically, the filter generation fully exploits the perception of the object context and scale information, making the decoding process more discriminatively. In general, the whole network architecture is composed of three parts, i.e., a feature extraction network, feature transfer modules, and a dynamic decoder network.

1) Feature Extraction Network: We adopt the VGG-16 network [35] as the feature extraction network in our proposed method. The last pooling layer along with the layers after the fifth block is removed and employed for feature extraction. For simplicity, the outputs of the five blocks are denoted as E_i, where $i \in \{1, 2, 3, 4, 5\}$ is the block index. It is noteworthy that E_i corresponds to the feature map of the last convolutional layer in each block, i.e., conv1-2, conv2-2, conv3-3, conv4-3, and conv5-3.

2) Dynamic Transfer Attention: The DTA module is used for interference control between the encoder and the decoder. DTA first generates an attention map and then uses it to filtrate encoder features selectively. The generated attention map indicates the importance of each spatial position as well as each feature channel, thus, can enhance feature representations and suppress interference from the encoder flexibly and comprehensively. It thoroughly considers three types of context, i.e., local spatial context, global spatial context, and channel context. The calculation process of DTA is simple and intuitive, which will be illustrated in Section III-B. We can briefly express DTA by the following function:

$$T_i = \mathcal{F}(E_i) \tag{1}$$

where T_i denotes the output of the DTA module, it has the same spatial sizes and channels as E_i. The effectiveness of DTA is examined in Section IV-C.

3) Pyramid Scene- and Location-sensitive Dynamic Filtering Module: The PSLD filtering module is deployed at each level of the decoder network, targeting inferring salient object regions from the filtrated feature representations. Different from existing decoder networks or modules that consist of convolution layers with fixed parameters, PSLD can adjust its filtering parameters as scenes and locations change. This increases the adaptability of the proposed network to diverse and complex scenarios in optical RSIs. Furthermore, to leverage rich sample-specific context for dynamic filter generation and saliency computation, PSLD adopts a pyramid structure to capture multiscale context information. Suppose we denote the function of the PSLD module as $\mathcal{H}$, the decoder process can be expressed as

$$\tilde{D}_{i+1} = \mathrm{Up}(\mathrm{Conv}_{1\times1}(D_{i+1})) \tag{2}$$

$$D_i = \mathcal{H}(\mathrm{Cat}(\mathrm{Conv}_{1\times1}(T_i), \tilde{D}_{i+1})) \tag{3}$$

where Up denotes the upsampling layer which has a factor of 2, $\mathrm{Conv}_{1\times1}$ denotes a 1×1 convolution layer followed by batch normalization and ReLU layers, and Cat concatenates the inputs along the channel axis. $i \in \{1, 2, 3, 4, 5\}$ denote layer indexes, where the fifth layer is without upsampling and concatenation operation. The details of PSLD will be illustrated in Section III-C and its effectiveness will be examined in Section IV-C.

B. Dynamic Transfer Attention Module

Transferring informative features from the encoder to the decoder plays an important part in generating high-quality saliency maps. However, existing encoder–decoder-based models for RSI-SOD ignore the interference control between the encoder and the decoder. Thus, the features are not filtrated properly for saliency calculation, which may introduce cluttered and noisy context information. To mitigate this issue and improve the model adaptability, we propose a DTA module to selectively activate feature responses and transfer them to the decoder network.

The explanatory workflow of DTA is depicted in Fig. 3. DTA generates attention for each spatial location as well as each channel in feature maps. To generate a comprehensive

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attention map, DTA leverages three types of context information, i.e., local spatial context, global spatial context, and channel context. DTA obtains the full-tensor contextual information in attention calculation; however, it consumes much less computational overhead than the conventional self-attention operation [48], [49].

All three types of attention mechanisms are calculated by convolution layers. The local spatial context-based attention is implemented by a standard local convolution, the global spatial context-based attention by a dilated convolution with a large kernel size, and the channel context-based attention by a standard 1×1 convolution. The calculation process can be expressed as follows:

$$A_i^l = \text{Conv}(E_i; k_l, d_l, g_l) \tag{4}$$
$$A_i^g = \text{Conv}\left(A_i^l; k_g, d_g, g_g\right) \tag{5}$$
$$A_i^c = \text{Conv}\left(A_i^g; k_c, d_c, g_c\right) \tag{6}$$
$$T_i = A_i^c * E_i \tag{7}$$

where Conv denotes the convolution layer, and $\{k_x, d_x, g_x | x \in (l, g, c)\}$ denote kernel size, dilation rate, group number of a convolution layer. E_i is the encoder feature at ith layer, A_i^l, A_i^g, and A_i^c denote outputs of local spatial convolution, global spatial convolution, and channel convolution, respectively. The symbol $*$ denotes elementwise multiplication. T_i denotes the final output of DTA. It should be noted that the input channels are equal to the output channels in the Conv setting. The final attention map A_i^c indicates the importance of each spatial location as well as each channel in the feature map, and thus, can be used for filtering the encoder feature before transferring it to the decoder. Specifically, to properly model the global spatial context, we set the kernel size k_g adaptive to the layer index i. While deeper layers need smaller kernel sizes to capture global context information, shallower layers would need larger kernel sizes to capture enough global context information. The layer-adaptive setting is superior to the fixed kernel size setting, and its effectiveness is verified in Section IV.

We set the hyperparameters of Conv layers in (4)–(6) as follows. For generating local spatial attention A_i^l, we set the hyperparameters of the convolution layer to: $k_l = 5, d_l = 1$, and $g_l = 1$. For generating global spatial attention A_i^g, we set the hyperparameters of the convolution layer to: $k_g = \varphi(i), d_g = 3$, and $g_g = 1$. For generating channel attention A_i^c, we set the hyperparameters of the convolution layer to: $k_c = 1, d_c = 1$, and $g_c = 1$. k_g is set adaptive to the layer index, which is empirically studied and verified in Section IV. Applying (4)–(7) to the encoded features of each layer $E_i, i \in \{1, 2, 3, 4, 5\}$, we obtain the activated feature outputs $S_i, i \in \{1, 2, 3, 4, 5\}$. Then, they are transferred to the decoder network for saliency computation.

C. Pyramid Scene- and Location-Sensitive Dynamic Filtering Module

The PSLD filtering module is the key component of our proposed architecture. It targets dynamically generating convolution kernels that are sensitive to the scene-level and location-level context of specific samples and employing them for calculating saliency. The detailed pipeline of PSLD is presented in Fig. 4. In order to exploit rich sample-specific context in the dynamic decoding process, PSLD adopts a pyramid structure inspired by previous works on image segmentation [50], [51]. PSLD is composed of four subbranches, each performing dynamic filtering with a distinctive dilation rate. Each branch input feature is fed into an SLD filtering module to produce the output feature representation.

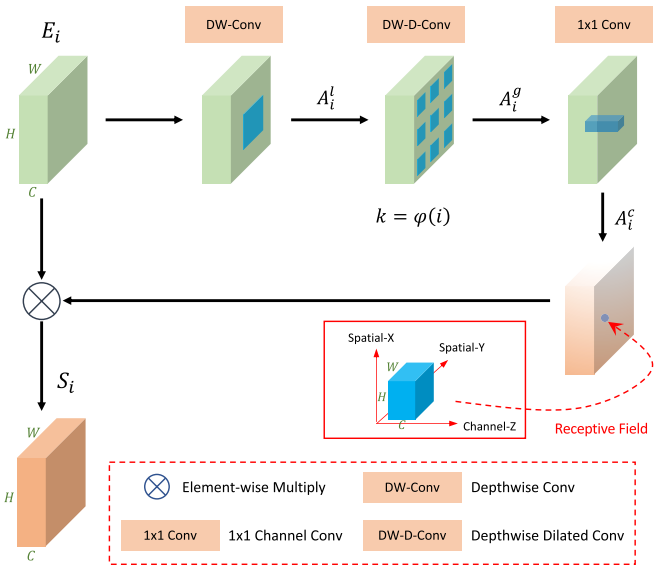


Fig. 3. Workflow of the DTA module.

Suppose the input feature is denoted as F. It is first fed into a convolution layer for feature transformation and then split into four groups evenly along the channel dimension

$$M = \text{Conv}_{1\times1}(F) \tag{8}$$
$$M^1, M^2, M^3, M^4 = \text{Split}(M). \tag{9}$$

After that, each split feature is processed dynamically by an SLD module, generating output $\tilde{M}^i, i \in \{1, 2, 3, 4\}$. To further enhance cross-scale feature interactions, we add residual-like operations between the four SLD modules. The output of each SLD can be expressed as

$$\tilde{M}^i = \begin{cases} \mathcal{SLD}(M^i), & i = 1 \\ \mathcal{SLD}(M^i + \tilde{M}^{i-1}), & i = 2, 3, 4. \end{cases} \tag{10}$$

The final output of PSLD is calculated by

$$D = \text{Conv}_{1\times1}(\text{Cat}(\tilde{M}^1, \tilde{M}^2, \tilde{M}^3, \tilde{M}^4, \tilde{M}^o)) \tag{11}$$

where Cat denotes concatenation along channel dimension, $\tilde{M}^o$ is derived by feeding input feature into a standard 3×3 convolution layer, compressing channels to the same as each split feature map. The $\tilde{M}^o$ connection path is omitted in the figure.

In the following, we elaborate on SLD in detail. As shown in the right-hand side part of Fig. 4, SLD consists of two filter generation flows, i.e., a scene-sensitive dynamic filter and a location-sensitive dynamic filter. The former is conditioned on the global scene-aware representation, while the latter

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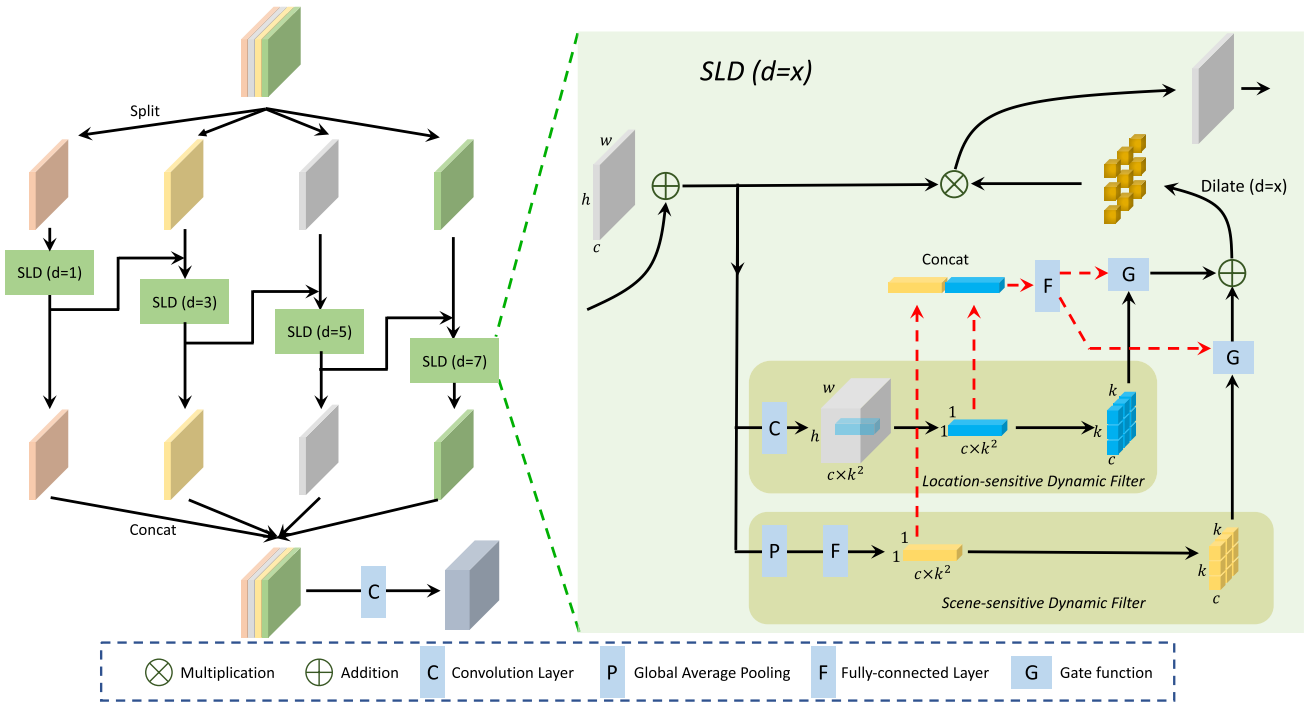


Fig. 4. Illustration of the PSLD filtering module. The overall pipeline of PSLD is shown in the left part. PSLD consists of four subbranches in parallel, each with a SLD filtering module. The detailed pipeline of SLD is shown in the right-hand side part.

is conditioned on location-dependent representation. A gate attention mechanism is used to weigh the importance of scene- and location-dependent contexts, combining the two types of filters into the final dynamic filter.

1) Scene-Sensitive Dynamic Filter: Scene-sensitive dynamic filtering aims to decode the feature representation by the global-view-based dynamic kernels. Given the input $M \in \mathbb{R}^{h\times w\times c}$ (we omit the split superscript for simplicity), we apply global average pooling (GAP) to derive a global descriptor. Then, we feed it through a fully connected layer (implemented by 1×1 convolution), obtaining an output vector that is used as the filter parameters. The filter parameter generation process can be expressed as

$$P_s = \text{Conv}_{1\times1}(\text{GAP}(M)) \tag{12}$$

where GAP denotes global average pooling layer. The $\text{Conv}_{1\times1}$ operates on the global descriptor and expands the channel dimension to $c \times k^2$, thus obtaining $P_s \in R^{1\times1\times(c\times k^2)}$. The scene-sensitive dynamic convolution kernel can be obtained by reshaping P_s to $K_s \in \mathbb{R}^{k\times k\times c}$. It should be noted that the scene-sensitive dynamic kernel is applied to all locations equally, while the location-sensitive dynamic kernel is generated per pixel location independently.

2) Location-Sensitive Dynamic Filter: Location-sensitive dynamic filtering aims to decode the feature representation by location-dependent dynamic kernels. A standard 3×3 convolution layer is applied to the input feature M, expanding channel dimension by a factor of k^2

$$P_l = \text{Conv}_{3\times3}(M) \tag{13}$$

where $P_l \in \mathbb{R}^{h\times w\times(c\times k^2)}$. Thus, each spatial location (p, q) can be extracted from P_l, deriving $P_l^{(p,q)} \in \mathbb{R}^{1\times1\times(c\times k^2)}$. The dynamic convolution kernel at that location can be obtained as $K_l^{(p,q)} \in \mathbb{R}^{c\times k\times k}$.

3) Gate Attention: We adopt a gate attention mechanism to determine the importance of scene-dependent and location-dependent contextual information across space when generating the final dynamic filters. To this end, scene-sensitive and location-sensitive dynamic filter parameters are concatenated and then fed into a fully connected layer (implemented by 1×1 convolution) to generate gate attention values

$$G = \sigma\left(\text{Conv}_{1\times1}\left(\text{Cat}\left(P_s, P_l^{(p,q)}\right)\right)\right) \tag{14}$$

where Cat denotes concatenation along channel dimension, and σ denotes sigmoid function. The output channel of $\text{Conv}_{1\times1}$ is set to 2, each corresponding to one of the gate attention values for the scene-sensitive filter and location-sensitive filter, respectively. Let $G_s, G_l \in [0, 1]$ denote the scalar values of the scene-dependent gate and the location-dependent gate, respectively, which are extracted from $G \in \mathbb{R}^{1\times1\times2}$. The final dynamic filter is generated by selecting the relevant filters based on attention gates

$$K^{(p,q)} = G_s * K_s + G_l * K_l^{(p,q)} \tag{15}$$

where $K^{(p,q)} \in \mathbb{R}^{k\times k\times c}$ is the final dynamic filter kernel at spatial location (p, q) of the input feature representation.

4) Dynamic Filtering Process: Input feature M is convolved depthwise by the dynamic convolution kernels to generate output features. Specifically, the dynamic kernel is enlarged by a certain dilation rate d to cover a square neighborhood $[(k-1)d+1, (k-1)d+1]$. Dilated convolution [52] is a widely used technique that expands the kernel size by inserting spaces (dilation rate) between its consecutive elements. With the dilation rate, the dynamic kernel can cover a wider field of view without increasing the

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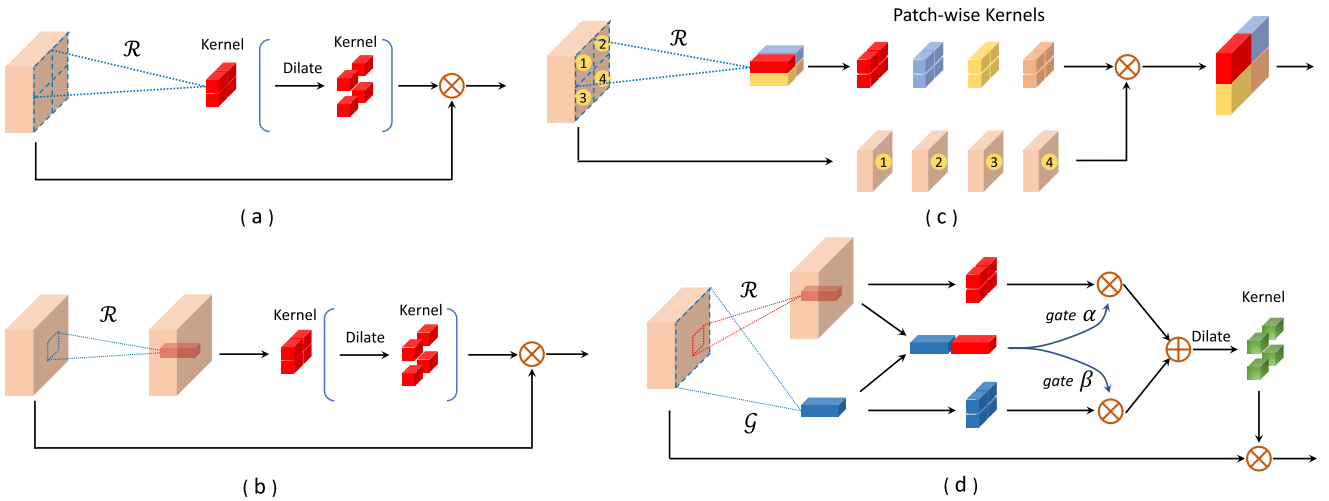


Fig. 5. Comparison of different dynamic convolution pipelines adopted in dense-pixel prediction. (a) Single-dynamic kernel [33], [42]. (b) Locationwise dynamic kernels [34]. (c) Patchwise dynamic kernels [43], [44]. (d) Our proposed SLD kernels. The bracket denotes optional dilation operation. $\mathcal{R}$ denotes regional context mapping, while $\mathcal{G}$ denotes global context mapping.

computational cost. The output value of a unit at the position (p, q, c), denoted as $\tilde{M}_{p,q,c}$, is given by

$$\tilde{M}_{p,q,c} = \sum_{(i,j)\in N} K^{(p,q)}_{(i,j,c)} * M_{(p+d*i,q+d*j,c)} \tag{16}$$

where $N = \{(-\lceil (k-1)/2 \rceil, -\lceil (k-1)/2 \rceil), \ldots, (\lceil k/2 \rceil, \lceil k/2 \rceil)\}$ denote the range of position indexes in the kernel, d denotes the dilation rate when performing convolution. The dilation rates of the four SLD branches are set to 1, 3, 5, and 7, respectively to decode saliency representations at multiple scales. Finally, the representation is updated by a residual connection from M to obtain the output of the SLD module.

D. Loss Function

This section elaborates on our loss function for optimizing the proposed model. We incorporate two types of loss functions for supervising the decoder outputs at the pixel level. The first loss is the commonly used binary cross entropy (BCE), which can be expressed as

$$\mathcal{L}_{\text{bce}}(S, G) = -\sum_{i=1}^{H}\sum_{j=1}^{W} G_{i,j} \log S_{i,j} + (1 - G_{i,j}) \log(1 - S_{i,j}) \tag{17}$$

where S and G denote the predicted saliency map and the ground truth. The second loss is the IoU function, which is originally proposed for measuring the similarity of two sets and used as a standard evaluation measure for object detection. The IoU loss is calculated by

$$\mathcal{L}_{\text{iou}}(S, G) = 1 - \frac{\sum_{i=1}^{H}\sum_{j=1}^{W} S_{i,j} G_{i,j}}{\sum_{i=1}^{H}\sum_{j=1}^{W} (S_{i,j} + G_{i,j} - S_{i,j} G_{i,j})}. \tag{18}$$

Finally, the total loss of our proposed method can be expressed as

$$\mathcal{L} = \sum_{l=1}^{5} [\mathcal{L}_{\text{bce}}(S^l, G) + \mathcal{L}_{\text{iou}}(S^l, G)] \tag{19}$$

where l denotes the layer index. It is noted that S^l is obtained by feeding the decoding feature of each layer into a 1×1 convolution layer and upsampling to the same spatial size of the original input

$$S^l = \sigma(\text{Up}(\text{Conv}_{1\times 1}(D_i))) \tag{20}$$

where σ is the sigmoid function, Up is the bilinear upsampling layer that upsamples the predicted saliency map to the same spatial size as the input image. Five saliency maps are generated at the decoder layers during the training stage for model optimization. S^1 is viewed as the final saliency prediction of PCD-Net during the testing stage.

E. Comparison of Different Dynamic Convolution Pipelines

We present the comparison of different dynamic convolution pipelines adopted in dense-pixel prediction tasks (e.g., semantic segmentation, SOD, and so on) in Fig. 5. Current dynamic convolution pipelines can be categorized into several classes in terms of dynamic filter generation: 1) a global dynamic kernel [as shown in Fig. 5(a)]. 2) Location-specific dynamic kernels [as shown in Fig. 5(b)]. 3) Patchwise dynamic kernels [as shown in Fig. 5(c)]. The first pipeline generates a single-dynamic kernel and applies it to all spatial locations equally. The second pipeline performs convolution at each spatial location with a generated location-specific dynamic kernel. The third pipeline leverages patch-level contexts to generate patchwise dynamic kernels and perform convolution at each patch separately. Existing methods leverage on regional context to generate dynamic filter kernels, e.g., reference [33] uses regional pooling operation to generate a global dynamic kernel and reference [34] employs a local convolution layer to generate location-specific kernels. However, they ignore the role of global scene-level contextual information in dynamic kernel generation. Fig. 5(d) shows the pipeline of our proposed dynamic convolution. It mainly consists of three successive steps: first, it generates a scene-sensitive and a location-sensitive dynamic kernel based on the global and local contextual information, respectively; second, a gate attention

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mechanism is applied to combine them selectively into a final dynamic kernel; and third, the dynamic kernel is dilated and convolved with the input representations to derive output. The primary distinctiveness of the proposed dynamic convolution pipeline can be summarized as follows: it utilizes scene-level global context in the kernel generation process in addition to region-level context information; it applies second-order attention to adjust the kernel based on weighing the importance between location-specific and scene-dependent contexts.

IV. Experiment

A. Experiment Protocol

1) Datasets: We evaluate our model on two public RSI-SOD benchmarks: ORSSD [21] and EORSSD [22]. ORSSD contains 800 optical RSIs with pixel-level annotations, and it is divided into 600 training samples and 200 testing samples. EORSSD is the expanded version of ORSSD, which contains 2000 optical sensing images. It is divided into 1400 training samples and 600 test samples.

2) Evaluation Metrics: We adopt five widely used evaluation metrics to evaluate our method and other compared methods, including S-measure [53], F-measure [6], E-measure [54], mean absolute error (MAE), and precision-recall (PR) curve. F-measure is the weighted average of precision and recall, which is computed by

$$F_\beta = \frac{(1+\beta^2)\text{Precision} \times \text{Recall}}{\beta^2 \times \text{Precision} + \text{Recall}} \tag{21}$$

where β^2 is set to 0.3 to give greater weight to precision in SOD [10]. S-measure measures object-aware and region-aware structural similarity between the predictions and GTs. It can be expressed as

$$S_m = \alpha * S_o + (1-\alpha) * S_r \tag{22}$$

where S_o and S_r denote object-aware and region-aware structural similarity, α is set to 0.5 for assigning equal contribution to both similarities [53]. E-measure evaluates the similarity between the predicted saliency maps and the GT by considering the local- and image-level saliency values simultaneously

$$E_\phi = \frac{1}{W \times H} \sum_{i=1}^{W} \sum_{j=1}^{H} \theta(\phi) \tag{23}$$

where ϕ is the alignment matrix and $\theta(\phi)$ denotes the enhanced alignment matrix [54]. MAE measures the average absolute difference at all pixel locations. PR curve presents the precision rate against the recall rate by varying the threshold of the saliency map from 0 to 255.

3) Implementations: Adam optimizer [55] is adopted as the model optimizer for parameter updating. The initial learning rate is set to 5×10^{-5}, and the total number of training epochs is set to 40. All the experiments are conducted on a GTX 3090 GPU. The input image is resized to 256×256 before feeding into the proposed model for both training and testing. Following [22] and [24], we train our model on the EORSSD dataset and test it on the test sets of the EORSSD and ORSSD datasets. The training data are augmented by conducting rotation with 90°, 180°, and 270° and mirror reflection, acquiring seven additional samples for each image.

B. Comparison With Other Methods

We compare our method with 18 state-of-the-art methods, including six CNN-based methods for NSIs: DSS [10], R3Net [56], EGNet [15], RADF [57], GateNet [11], PoolNet [14], three traditional methods for RSIs: SMFF [20], CMC [58], VOS [59], nine CNN-based methods for RSIs: LVNet [21], PDFNet [23], EMFINet [25], MJRBM [28], DAFNet [22], FSMINet [24], MCCNet [60], SARNet [26], and ERPNet [29]. The saliency maps of the compared methods are provided by the authors or generated using publicly source codes. In the following, we conduct comprehensive evaluations in terms of quantitative comparison, qualitative comparison, and computational efficiency comparison.

1) Quantitative Comparison: Table I reports the comparison of our proposed models against state-of-the-art models on eight evaluation metrics. These models are split into three groups from top to bottom: CNN-based NSI-SOD methods, traditional RSI-SOD methods, and CNN-based RSI-SOD methods. We can observe from Table I that the CNN-based NSI-SOD methods perform better than the traditional hand-designed RSI-SOD methods, while generally lagging behind the CNN-based RSI-SOD methods. Specialized methods for RSIs, e.g., ERPNet and MCCNet, achieve much better performance than NSI-SOD methods, which demonstrates the necessity of designing exclusive models for RSI-SOD. On the ORSSD dataset, our method ranks first in eight metrics, showing superiority to other compared methods. On the EORSSD dataset, our method achieves a competitive performance and is still among the top methods. It ranks first in one metric, second in five metrics, and third in two metrics. It reaches comparable scores to MCCNet and EMFINet, which further utilizes edge cues as auxiliary information. It is noteworthy that our method performs significantly better than the second-best method, especially on the structural measure S_m (0.9444 against 0.9376 on ORSSD and 0.9346 against 0.9317 on EORSSD). Fig. 6 presents the comparison of the PR curve and F-measure curve. It can be seen that our method is better than other compared methods in general.

2) Qualitative Comparison: We conduct a visual comparison of our method against other methods in Fig. 7, which shows some representative scenarios in RSIs. *1) Salient objects with complex and cluttered backgrounds*—Most NSI-SOD methods, e.g., DSS and RADF, have difficulty in dealing with cluttered backgrounds. RSI-SOD methods can better eliminate background clutters; however, they may falsely highlight some confusing background regions (as shown in the first row) or regard salient objects as background (as shown in the third row). In comparison, our method can accurately highlight all salient objects while suppressing cluttered backgrounds. *2) Salient objects with large area occupation*—Salient objects with overwhelmingly large sizes usually appear in RSIs. Our method can uniformly highlight the whole salient regions as well as preserve clear object boundaries. In contrast, other

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TABLE I

QUANTITATIVE EVALUATION. THE EVALUATION METRICS INCLUDE: MAE (M), STRUCTURE MEASURE (S_m), MAXIMUM, MEAN, AND ADAPTIVE F-MEASURE ($F_\beta^{\max}$, F_β^{mean}, AND F_β^{ada}), MAXIMUM, MEAN, AND ADAPTIVE E-MEASURE ($E_\varepsilon^{\max}$, $E_\varepsilon^{\text{mean}}$, AND $E_\varepsilon^{\text{ada}}$). FOR MAE SMALLER IS BETTER, WHILE FOR THE OTHER METRICS LARGER IS BETTER. THE BEST THREE RESULTS ARE MARKED IN RED, GREEN, AND BLUE, RESPECTIVELY

Method	EORSSD								ORSSD							
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$F_\beta^{\text{mean}}\uparrow$	$F_\beta^{\text{ada}}\uparrow$	$E_\varepsilon^{\max}\uparrow$	$E_\varepsilon^{\text{mean}}\uparrow$	$E_\varepsilon^{\text{ada}}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$F_\beta^{\text{mean}}\uparrow$	$F_\beta^{\text{ada}}\uparrow$	$E_\varepsilon^{\max}\uparrow$	$E_\varepsilon^{\text{mean}}\uparrow$	$E_\varepsilon^{\text{ada}}\uparrow$
DSS	.7868	.0186	.6849	.5801	.4597	.9186	7631	.6933	.8262	.0363	.7467	.6962	.6202	.8860	.8362	.8085
R3Net	.8184	.0171	.7498	.6302	.4165	.9483	.8294	.6462	.8141	.0399	.7456	.7383	.7379	.8913	.8681	.8887
EGNet	.8601	.0110	.7880	.6967	.5379	.9570	.8775	.7566	.8721	.0216	.8332	.7500	.6452	.9731	.9013	.8226
RADF	.8179	.0168	.7446	.6582	.4933	.9140	.8567	.7261	.8259	.0382	.7619	.6856	.5730	.9130	.8298	.7678
GateNet	.9114	.0095	.8566	.8228	.7109	.9610	.9385	.8909	.9186	.0137	.8871	.8679	.8229	.9664	.9538	.9428
PoolNet	.8207	.0210	.7545	.6406	.4611	.9292	.8193	.6836	.8403	.0358	.7706	.6999	.6166	.9343	.8650	.8124
VOS	.5082	.2096	.2765	.2107	.1836	.5982	.4886	.4767	.5366	.2151	.3471	.2717	.2633	.6514	.5352	.5826
SMFF	.5401	.1434	.5176	.2992	.2083	.7744	.5197	.5014	.5312	.1854	.4417	.2684	.2496	.7402	.4920	.5676
CMC	.5798	.1057	.3268	.2692	.2007	.6803	.5894	.4890	.6033	.1267	.3913	.3454	.3108	.7064	.6417	.5996
LVNet	.8630	.0146	.7794	.7328	.6284	.9254	.8801	.8445	.8815	.0207	.8263	.7995	.7506	.9456	.9259	.9195
PDFNet	-	-	-	-	-	-	-	-	.9112	.0149	.8726	.8435	.7924	.9608	.9403	.9341
EMFINet	.9290	.0084	.8720	.8486	.7984	.9711	.9604	.9501	.9366	.0109	.9002	.8856	.8617	.9737	.9671	.9663
MJRBM	.9197	.0099	.8656	.8239	.7066	.9646	.9350	.8897	.9204	.0163	.8842	.8566	.8022	.9623	.9415	.9328
DAFNet	.9166	.0060	.8614	.7845	.6427	.9861	.9291	.8446	.9191	.0113	.8928	.8511	.7876	.9771	.9539	.9360
FSMINet	.9255	.0079	.8678	.8436	.8015	.9666	.9567	.9490	.9361	.0101	.9041	.8878	.8710	.9759	.9672	.9693
MCCNet	.9317	.0074	.8842	.8618	.8238	.9733	.9679	.9557	.9376	.0109	.9053	.8933	.8795	.9723	.9682	.9660
SARNet	.9240	.0099	.8719	.8541	.8304	.9620	.9555	.9536	.9134	.0187	.8850	.8619	.8512	.9557	.9477	.9464
ERPNet	.9210	.0089	.8632	.8304	.7554	.9603	.9401	.9228	.9254	.0135	.8974	.8745	.8356	.9710	.9566	.9520
OURS	.9346	.0069	.8830	.8590	.8088	.9744	.9645	.9507	.9444	.0094	.9138	.9014	.8839	.9809	.9742	.9731

TABLE II

COMPARISON ON MODEL COMPLEXITY

Method	Params(M)↓	FLOPs(G)↓	FPS↑	Input Res
GateNet	100.0	96.3	**51.5**	256*256
EGNet	108.1	313.9	43.53	256*256
PoolNet	52.5	97.6	80.9	256*256
LVNet	207.0	-	1.5	128*128
PDFNet	228.0	-	-	128*128
MJRBM	43.8	101.3	15.5	256*256
EMFINet	95.1	353.1	18.6	256*256
FSMINet	**3.6**	**9.7**	17.7	256*256
DAFNet	29.4	548.1	42.5	256*256
MCCNet	67.7	234.2	39.5	256*256
ERPNet	77.2	343.2	32.5	256*256
SARNet	25.9	137.3	43.4	256*256
OURS	17.7	52.9	30.2	256*256

methods may discard some salient object parts (e.g., ERPNet and SARNet), or derive blurry boundaries (e.g., LVNet and EMFINe). *3) Multiple small/tiny salient objects*—When there are multiple small/tiny salient objects presented in an RSI, current methods tend to have trouble extracting complete salient regions (e.g., DAFNet and ERPNet). Due to the strong scale adaptability, our model shows robust performance in detecting multiple small/tiny salient objects. *4) Salient objects with low contrast*—Salient objects with low contrast appear to be a challenging scenario in RSIs. Our method can still achieve promising results in such scenarios as presented in rows 10–12 of Fig. 7. *5) Salient objects with irregular topology*—Salient objects in RSIs, such as rivers, roads, and so on, often appear with irregular topology. Existing methods have trouble handling such cases due to the lack of adaptability to rarely seen topologies. In comparison, our method can adapt well to complex topologies and highlight the whole salient regions.

3) Computational Efficiency Analysis: We measure the computational complexity using three metrics, including network parameters, floating-point operations (FLOPs), and frames per second (FPS). The evaluation results are reported in Table II. We can observe that our model has a much lower amount of network parameters and FLOPs than most of the compared methods. Since FSMINet is designed specifically for the lightweight purpose, it has the lowest network parameters and FLOPs. Our model has a total number of parameters 17**M**, in which the backbone network occupies 14.6**M** and the proposed novel modules (including DTA and PSLD) along with the classification layers take up around 3**M**. In addition, our method achieves a real-time inference speed of 30.2 FPS. The evaluations demonstrate that our model is advantageous in computational efficiency compared with other well-performing methods.

C. Ablation Study

1) Contribution of Key Modules in PCD-Net: To measure the contributions of DTA and PSLD modules in PCD-Net, we conduct ablation experiments and report them in Table III. First of all, we construct a baseline (BL) model which is PCD-Net with DTA and PSLD removed ["BL"]. Then, we add DTA to the BL model ("BL + DTA") and PSLD to the BL model ("BL + PSLD"), respectively. Finally, we add both of the two modules to the BL model, leading to the final proposed architecture ("BL + PSLD + DTA").

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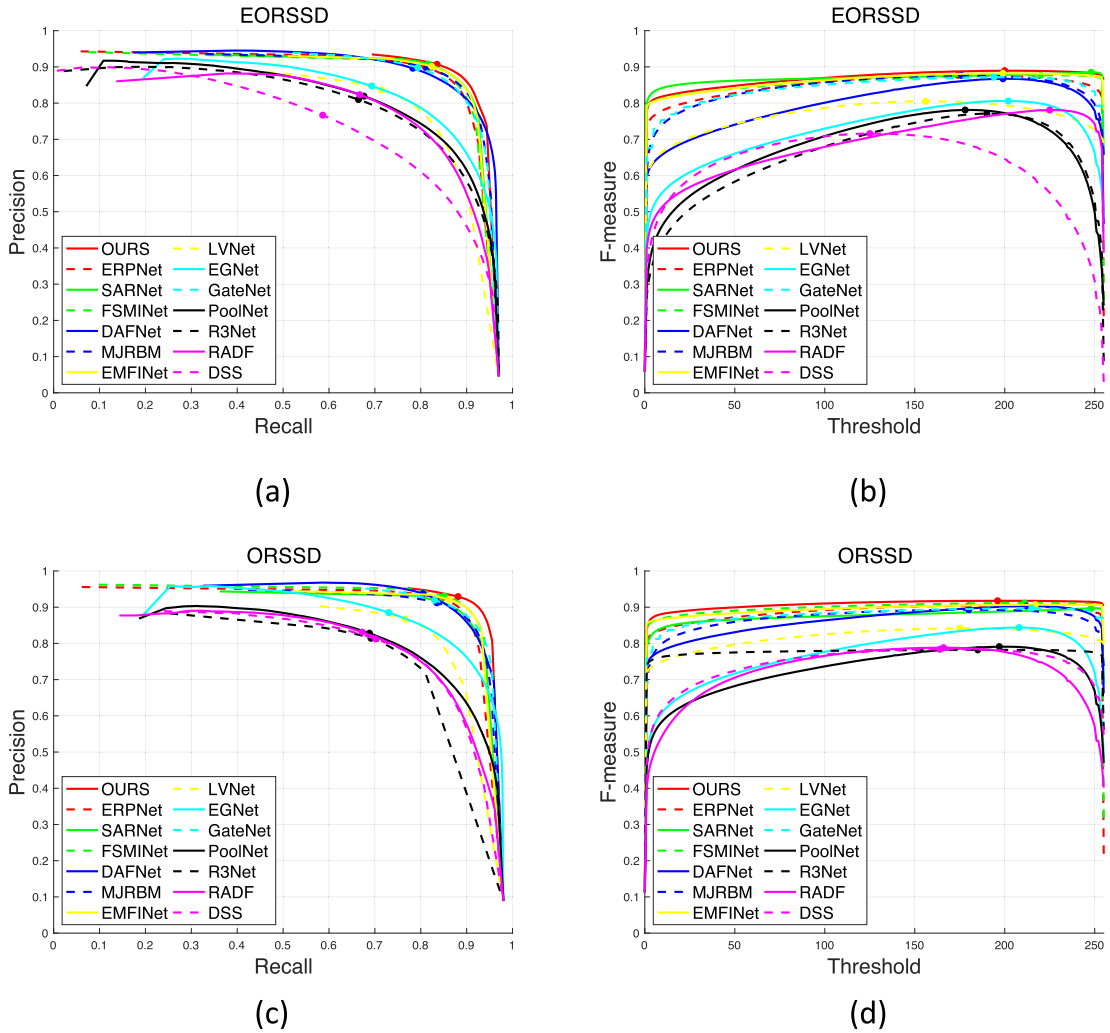


Fig. 6. Quantitative evaluation of different methods in terms of PR curve and F-measure curve. (a) PR curves on EORSSD. (b) PR curves on EORSSD. (c) F-measure curves on ORSSD. (d) F-measure curves on ORSSD.

TABLE III
ABLATION STUDY ON THE PROPOSED KEY COMPONENT OF SLDF-NET

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
Baseline (BL)	.9130	.0199	.8699	.9131	.0116	.8497
BL+DTA	.9286	.0139	.8935	.9285	.0081	.8763
BL+PSLD	.9395	.0106	.9072	.9317	.0083	.8818
BL+PSLD+DTA	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**

On the EORSSD dataset, the BL model achieves 0.9131 S_m, 0.116 MAE, and 0.8497 $F_\beta^{\max}$. DTA improves the performance of BL by 1.69% on S_m, 0.0035 on MAE, and 2.82% on $F_\beta^{\max}$, respectively. PSLD improves the performance of BL by 2.04% on S_m, 0.0033 on MAE, and 3.78% on $F_\beta^{\max}$, respectively. The joint integration of the two modules improves the performance of BL by 2.35% on S_m, 0.0047 on MAE, and 3.92% on $F_\beta^{\max}$, respectively. A similar phenomenon can be observed in the ORSSD dataset. The complete model improves the BL from 0.9130 to 0.9444 on S_m (improve by 3.44%), from 0.0199 to 0.0094 on MAE (improve 0.0105), from 0.8699 to 0.9138 on $F_\beta^{\max}$ (improve by 5.05%), respectively. The corresponding visual ablation result is shown in Fig. 8. It can be inferred that the proposed modules improve the model performance significantly on RSI-SOD.

2) Effectiveness of Each Component in PSLD: PSLD is composed of two types of dynamically generated filters, i.e., location-dependent dynamic filter and scene-dependent dynamic filter. To explore the effectiveness of each dynamic filter and the complementarity between them, we conduct analyses and summarize the results in Table IV. The ablation is carried out by removing the location-dependent dynamic filter ("PSLD w/o. LDF") and scene-dependent dynamic filter ("PSLD w/o. SDF"), respectively. The effectiveness of gate attention is also examined in this study ("PSLD w/o. GA").

It can be observed that w/o. SDF and w/o. LDF are worse than the complete PSLD on two benchmarks. Specifically, on the ORSSD dataset, w/o. SDF drops the performance from 0.9444 to 0.9362 on S_m, from 0.0094 to 0.0120 on MAE, and from 0.9138 to 0.8989 on $F_\beta^{\max}$, while w/o. LDF drops the performance from 0.9444 to 0.9372 on S_m, from 0.0092 to 0.0108 on MAE, and from 0.9138 to 0.9072 on $F_\beta^{\max}$. Meanwhile, the same trend can be observed in the EORSSD dataset. It demonstrates that both scene-dependent filtering and location-dependent filtering are vital for the excellent performance. The ablation result of gate attention ("PSLD w/o. GA") verifies that the adaptive combination of scene-dependent and location-dependent filters is important for

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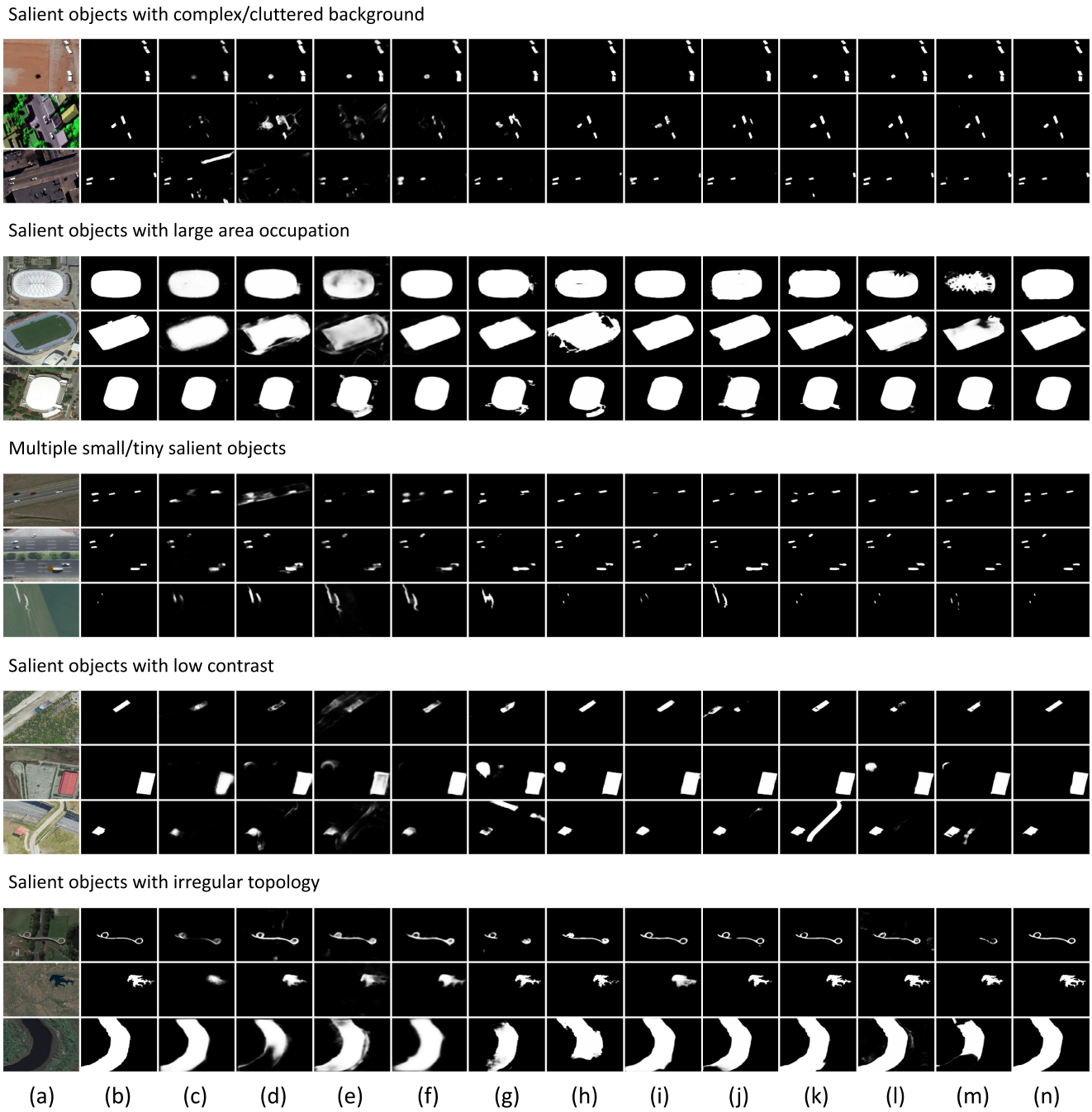


Fig. 7. Visual comparisons with 11 state-of-the-art methods, including four CNN-based NSI-SOD methods and seven CNN-based RSI-SOD methods. (a) Optical RSIs. (b) GT. (c) DSS. (d) RADF. (e) PoolNet. (f) EGNet. (g) LVNet. (h) EMFINet. (i) DAFNet. (j) FSMINet. (k) MCCNet. (l) SARNet. (m) ERPNet. (n) OURS.

modeling the accurate context information. We also examine the residual-like connections' direction between SLD modules in PSLD. It is observed that adding residual connections from SLD of small dilation rate to that of large dilation rate successively ("PSLD") performs better than adding connections in the opposite direction ["PSLD (L2S conn.)"] and adding no connections ["PSLD (no conn.)"].

The visual ablation result is shown in Fig. 9. It is easily observed that location-sensitive dynamic filter ("PSLD w/o. SDF") can capture more fine-grained details, e.g., object boundaries and potential small objects. The scene-sensitive dynamic filter ("PSLD w/o. LDF") tends to highlight the whole regions that are salient and connected. A combination of both filters is capable of extracting the salient regions uniformly with clear boundaries. Our method shows strong adaptability to the variation in object sizes and topologies, while other representative methods, e.g., EMFI, SARNet, and ERPNet, have trouble dealing with such large variations presented in RSIs.

3) Comparison of PSLD With Other Alternatives: We replace PSLD with alternative decoding modules for progressive multiscale saliency calculation, including a plain

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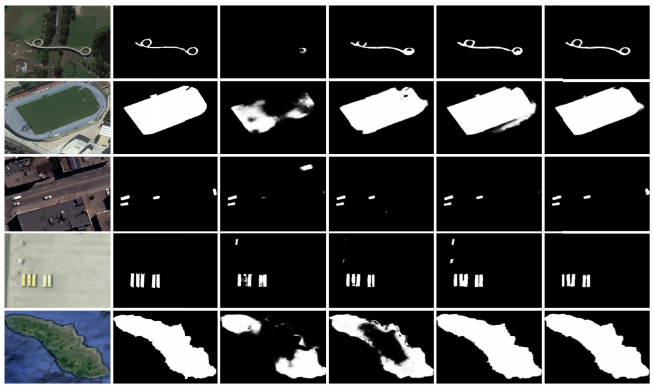


Fig. 8. Visual comparison of different variants of our proposed model. (a) Optical RSIs. (b) GT. (c) BL. (d) BL + DTA. (e) BL + PSLD. (f) BL + DTA + PSLD.

TABLE IV
ABLATION STUDY ON PSLD

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{\max}\uparrow$
PSLD	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**
PSLD (w/o. SDF)	.9362	.0120	.8989	.9325	.0075	.8818
PSLD (w/o. LDF)	.9372	.0108	.9072	.9329	.0072	.8820
PSLD (w/o. GA)	.9332	.0139	.8790	.9321	.0084	.8785
PSLD (no conn.)	.9403	.0104	.9062	.9337	.0074	.8816
PSLD (L2S conn.)	.9392	.0110	.9048	.9334	.0075	.8799
Plain Decoder	.9333	.0125	.8990	.9306	.0078	.8771
Res2Net_unit [61]	.9368	.0111	.9051	.9310	.0075	.8807
SCPC [62]	.9385	.0103	.9030	.9324	.0073	.8772

decoder, Res2Net block [61], and the SCPC module [62]. The plain decoder is obtained by replacing PSLD with a 3×3 standard convolution layer. Res2Net block conduct feature decoding by multiscale paralleled feature learning and cross-branch interaction. SCPC module is the extension of the Res2Net block, which adopts dilated convolutions in paralleled branches. The experimental results are shown in Table IV. We can find that the Res2Net block and SCPC perform better than the plain decoder, however, is worse than the proposed PSLD. To better comprehend the effectiveness of PSLD, we visualize the intermediate feature representations of the proposed model. We apply principal component analysis (PCA) to the last decoder layer feature D_1 for channel dimension reduction and retain the first five principal components for visualization. Fig. 10 presents the visualization comparison between our model with the plain decoder and with the PSLD module. We can observe that PSLD can help better suppress background clutters and extract salient regions with well-preserved boundaries. It is worth noting that salient objects are perceivable only in the first one or two principal components with the plain decoder, while they are distinguishable in almost all of the five principal components with PSLD.

4) Comparison of Different Dynamic Convolution Pipelines: We present the comparison result of different dynamic convolution pipelines (as illustrated in Section III-E) in Table V. To conduct a fair comparison, we simply replace

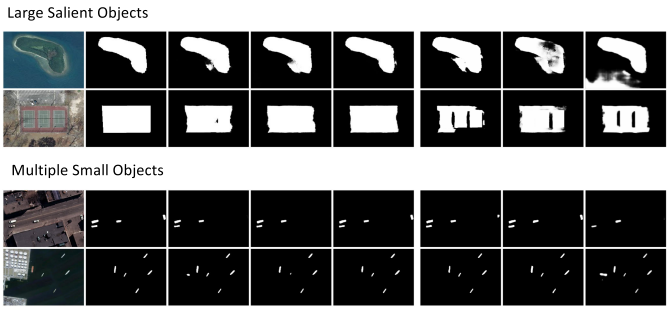


Fig. 9. Visual ablation study of the proposed PSLD. (a) Optical RSIs. (b) GT. (c) PSLD (w/o. SDF). (d) PSLD (w/o. LDF). (e) PSLD. (f) Saliency map generated by EMFI [25]. (g) Saliency map generated by SARNet [26]. (h) Saliency map generated by ERPNet [29].

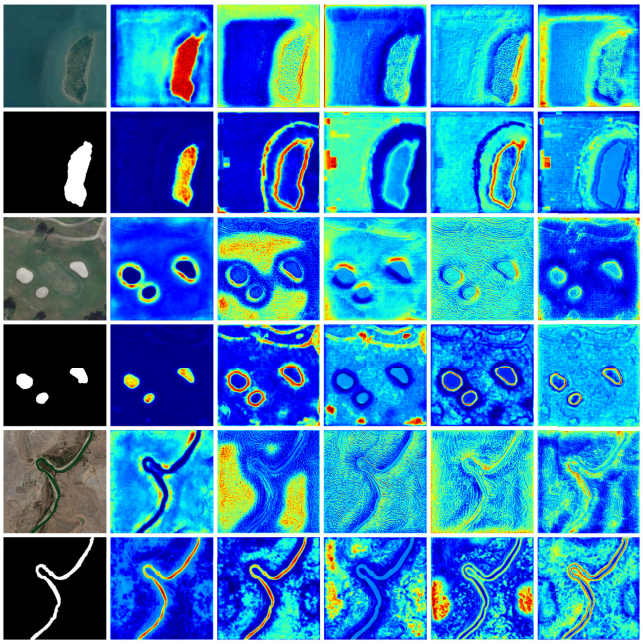


Fig. 10. PCA projection of the feature maps from the last decoder layer D_1. The odd rows show the visualization of our model with the plain decoder, while the even rows show the visualization of our model with PSLD. The first column shows the input RSIs and GTs. From the second to the sixth columns: PCA maps corresponding to the first to the fifth principal components in order.

the SLD module in the proposed model with other dynamic convolution pipelines, that is: 1) global dynamic convolution (GDC) [33]; 2) locationwise dynamic convolution (LDC) [34]; and 3) patchwise dynamic convolution (PDC) [43], as depicted in Fig. 5. It can be inferred from Table V that our proposed dynamic convolution pipeline achieves optimal results. It may be attributed to that our pipeline considers both scene-level and location-level contexts, and adjust their importance in a flexible way. In addition, 1) and 3) performs better than 2), demonstrating that regional context is more important than location-restrictive context in generating dynamic kernels.

5) Different Types of Attention Mechanisms in DTA: By default, our DTA module consists of three types of context aggregation operation, i.e., a spatial local context aggregation, a spatial long-range context aggregation, and a channel context aggregation. To validate the effectiveness of each component in DTA, we perform an ablation study by removing each

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TABLE V
COMPARISON OF DIFFERENT DYNAMIC CONVOLUTION PIPELINES

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
SLD	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**
(a) GDC	.9373	.0121	.9019	.9328	.0076	.8788
(b) LDC	.9339	.0126	.8971	.9305	.0079	.8767
(c) PDC	.9383	.0116	.9023	.9331	.0075	.8791

TABLE VI
ABLATION STUDY ON DTA MODULE

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
DTA	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**
DTA (w/o. LA)	.9342	.0138	.9021	.9298	.0089	.8770
DTA (w/o. GA)	.9369	.0120	.9023	.9317	.0086	.8796
DTA (w/o. CA)	.9387	.0115	.9052	.9325	.0081	.8808

TABLE VII
ABLATION STUDY ON KERNEL SIZE OF GLOBAL ATTENTION IN DTA

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
fix-(a)	.9372	.0118	.9026	**.9349**	.0072	.8824
adaptive-(a)	.9397	.0102	.9042	.9348	.0069	.8817
fix-(b)	.9366	.0120	.9024	.9324	.0079	.8793
adaptive-(b)	**.9444**	**.0094**	**.9138**	.9346	.0069	**.8830**
fix-(c)	.9375	.0112	.8988	.9344	**.0068**	.8794
adaptive-(c)	.9404	.0112	.9037	.9334	.0077	.8810

TABLE VIII
ABLATION STUDY ON THE LOSS FUNCTION

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
BCE	.9340	.0121	.9051	.9251	.0084	.8810
IoU	.9371	.0119	.9016	.9303	.0110	.8818
BCE & IoU	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**

component, respectively. The evaluation results are shown in Table VI. w/o. LA, w/o. GA, and w/o. CA denote removing local spatial context aggregation, long-range spatial context aggregation, and channel context aggregation, respectively. We can observe that removing each individual component will degrade the performance. For example, on the ORSSD dataset, w/o. LA, w/o. GA, and w/o. CA drop the performance from 0.9444 to 0.9342, 0.9369, and 0.9387 on S_m, from 0.0099 to 0.0138, 0.0120, and 0.0115 on MAE, from 0.9138 to 0.9021, 0.9023, and 0.9052 on $F_\beta^{\max}$, respectively. It also can be inferred that local spatial context has a more significant impact since w/o. LA degrades the performance more severely.

6) Receptive Fields of Global Attention in DTA: The global spatial attention in DTA is realized by a dilated convolution with a large kernel size. To explore the optimal kernel size for generating global spatial attention in DTA, we perform an ablation study and summarize the results in Table VII. We investigate two strategies for determining the kernel size: 1) a fixed kernel size for all levels; 2) an adaptive kernel size conditioned on specific layer levels. For the first strategy, we experiment with three different fixed kernel sizes, i.e., 7, 9, and 11. The three choices correspond to fix-(a), fix-(b), and fix-(c), respectively, in Table VII. For the second strategy, we experiment with adaptive kernels, which have larger sizes in shallower layers while having smaller sizes in deeper layers. This is because feature maps have a low resolution in deeper layers, while a high resolution in shallower layers. Thus, a small kernel size would capture the global receptive field for deeper layers, and a large kernel size would be necessary for shallower layers. To make a fair comparison, we keep the two strategies under the same computation budget. The corresponding second strategies are represented by adaptive-(a): {11, 9, 7, 5, 3}, adaptive-(b): {13, 11, 9, 7, 5}, and adaptive-(c): {15, 13, 11, 9, 7}. The numbers in the braces denote kernel sizes from the first (bottom) layer to the last (top) layer in the network. It is observed that DTA with adaptive kernel size always performs better than that with fixed kernel size, demonstrating that the layer-adaptive setting captures the global context more effectively. In particular, adaptive-(b) is best compatible with the backbone network and can model rich context dependencies across different layers.

7) Discussion of Loss Function: To validate the effectiveness of the hybrid loss function, we evaluate the performance of the training model with an individual loss (BCE loss or IoU loss only). The results are reported in Table VIII. It can be observed from Table VIII that our PCD-Net with only BCE loss or IoU loss can achieve a promising result, however, is worse than that of the hybrid loss. BCE provides supervision at the pixel level, while IoU emphasizes more on the detected region, and thus, both of them can contribute to the loss supervision in a complementary manner.

8) Scalability to Different Backbone Networks: Our method can be easily extended to other CNN encoder networks by replacing the original VGG-16 encoder. In Table IX, we replace VGG-16 with ResNet-50 [63] and MobileNet-v2 [64], and report their performance, respectively. We can observe that the performance of PCD-Net with ResNet-50 is worse than that with VGG-16. PCD-Net with MobileNet can still achieve promising results despite it having an extremely small number of parameters. Thus, it is suitable for applications on mobile devices.

9) Robustness to Noise Perturbation: We conduct experiments to examine the robustness of the proposed model to noise perturbation. We add Gaussian noise $\mathcal{N}(0, \sigma^2)$ to each pixel independently in the input image. The results are presented in Table X. It can be inferred that our model shows robustness to noise perturbation as the performance degrades slowly as the σ^2 increases. It can still achieve competitive performance even with a large noise (e.g., $\sigma^2 = 0.1$).

D. Comparison With Image Segmentation Methods

We compare the proposed method with four existing popular image segmentation methods, including two CNN-based methods: SFNet [65] and BiSeNetv2 [66] and two transformer-based methods: SETR [67] and SegFormer [68]. We set the output channel number of the segmentation models to 1 to generate a gray-scale saliency map. The training

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TABLE IX
PERFORMANCE EVALUATION ON OUR MODEL WITH DIFFERENT BACKBONE NETWORKS

Method	Size(M)	ORSSD			EORSSD		
		$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
PCD-Net (VGG)	17.8	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**
PCD-Net (ResNet)	41.9	.9353	.0116	.9033	.9291	.0081	.8772
PCD-Net (MobileNet)	1.7	.9296	.0122	.8895	.9251	.0080	.8696

TABLE X
PERFORMANCE OF THE PROPOSED METHOD ON DATA WITH GAUSSIAN NOISE

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
$\sigma^2=0$	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**
$\sigma^2=0.01$	.9336	.0107	.8988	.9227	.0077	.8678
$\sigma^2=0.05$	.9193	.0150	.8818	.9001	.0101	.8301
$\sigma^2=0.1$	.9086	.0151	.8634	.8856	.0116	.8132
$\sigma^2=0.2$	.8906	.0182	.8449	.8639	.0128	.7754
$\sigma^2=0.3$	.8736	.0230	.8192	.8384	.0158	.7390
$\sigma^2=0.4$	.8794	.0199	.8278	.8319	.0160	.7267

TABLE XI
COMPARISON WITH IMAGE SEGMENTATION METHODS

Method	ORSSD			EORSSD		
	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$	$S_m\uparrow$	$M\downarrow$	$F_\beta^{max}\uparrow$
PCD-Net	**.9444**	**.0094**	**.9138**	**.9346**	**.0069**	**.8830**
SegFormer	.9104	.0194	.8634	.8864	.0118	.8129
SETR	.9040	.0187	.8498	.8801	.0130	.7985
BiSeNetv2	.8853	.0189	.8495	.8776	.0126	.8074
SFNet	.9128	.0162	.8696	.8813	.0125	.8029

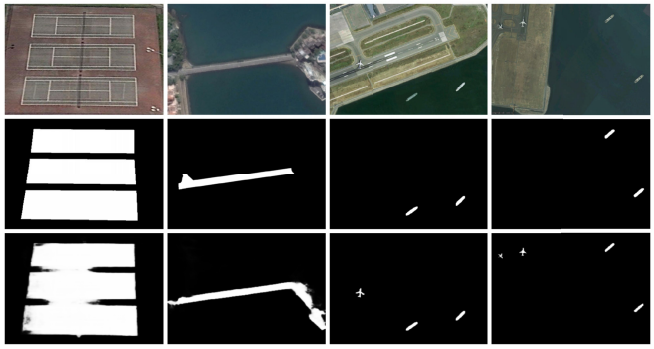


Fig. 11. Representative failure cases of the proposed PCD-Net. From top to bottom rows: optical RSIs, GT, and results produced by our model.

and testing are implemented in the experimental setting, as illustrated in Section. IV-A. The comparison results are reported in Table XI. It can be observed from Table XI that our method performs better than the compared image segmentation methods. Transformer-based methods fail to achieve ideal performance, possibly because of insufficient preservation of local context details, which are vital for SOD in remote sensing scenarios.

E. Discussion About Failure Cases

Our proposed PCD-Net can achieve state-of-the-art performance on RSI-SOD. However, since our architecture is simple and intuitive, there is still large room for further improvement. We present some representative failure cases in Fig. 11. It can be observed in case (1) that the proposed PCD-Net may fail in distinguishing object boundaries when salient objects are close to each other. Boundary cues may be utilized as auxiliary information for tackling such cases in the future. Case (2) shows that PCD-Net may treat connected regions as the extension of salient objects. Cases (3) and (4) show that PCD-Net can highlight extremely small objects (airplanes) in RSIs. This demonstrates that our model has a strong adaptability to extremely small targets. Though such small objects are not annotated in GTs, it is still confusing whether such extremely small objects belong to salient regions or not. In the future, we need to construct RSI-SOD datasets with a more careful annotation process to avoid such confusion.

V. CONCLUSION

In this article, we propose a PCD-Net for SOD in RSIs. The proposed PCD-Net is characterized by its strong adaptability to diversified variations in RSIs, such as object scales and complex topologies. Its adaptability is achieved mainly by two constituent modules: a PSLD filtering module and a DTA module. The former adaptively captures scene- and location-level context information and adjusts its filtering parameters for saliency calculation. The latter imposes a filtration mechanism between the encoder and the decoder, facilitating effective feature transfer. Extensive experimental evaluations on two public benchmarks demonstrate that the proposed PCD-Net can achieve state-of-the-art performance in RSI-SOD.

REFERENCES

[1] C. Guo and L. Zhang, "A novel multiresolution spatiotemporal saliency detection model and its applications in image and video compression," *IEEE Trans. Image Process.*, vol. 19, no. 1, pp. 185–198, Jan. 2010.

[2] S. Goferman, L. Zelnik-Manor, and A. Tal, "Context-aware saliency detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 10, pp. 1915–1926, Oct. 2012.

[3] W. Zhang, R. R. Martin, and H. Liu, "A saliency dispersion measure for improving saliency-based image quality metrics," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 28, no. 6, pp. 1462–1466, Jun. 2018.

[4] X. Wang, L. Ma, S. Kwong, and Y. Zhou, "Quaternion representation based visual saliency for stereoscopic image quality assessment," *Signal Process.*, vol. 145, pp. 202–213, Apr. 2018.

[5] Z. Ren, S. Gao, L.-T. Chia, and I. W. Tsang, "Region-based saliency detection and its application in object recognition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 24, no. 5, pp. 769–779, May 2014.

[6] R. Achanta, S. Hemami, F. Estrada, and S. Susstrunk, "Frequency-tuned salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1597–1604.

[7] M.-M. Cheng, N. J. Mitra, X. Huang, P. H. S. Torr, and S.-M. Hu, "Global contrast based salient region detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 3, pp. 569–582, Mar. 2015.

[8] J. Han, D. Zhang, X. Hu, L. Guo, J. Ren, and F. Wu, "Background prior-based salient object detection via deep reconstruction residual," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 25, no. 8, pp. 1309–1321, Aug. 2015.

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[9] J. Zhang and S. Sclaroff, "Saliency detection: A Boolean map approach," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 153–160.

[10] Q. Hou, M.-M. Cheng, X. Hu, A. Borji, Z. Tu, and P. H. S. Torr, "Deeply supervised salient object detection with short connections," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 4, pp. 815–828, Apr. 2019.

[11] X. Zhao, Y. Pang, L. Zhang, H. Lu, and L. Zhang, "Suppress and balance: A simple gated network for salient object detection," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 35–51.

[12] X. Gong, Z. Xie, Y. Liu, X. Shi, and Z. Zheng, "Deep salient feature based anti-noise transfer network for scene classification of remote sensing imagery," *Remote Sens.*, vol. 10, no. 3, p. 410, Mar. 2018.

[13] W. Diao, X. Sun, X. Zheng, F. Dou, H. Wang, and K. Fu, "Efficient saliency-based object detection in remote sensing images using deep belief networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 2, pp. 137–141, Feb. 2016.

[14] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3912–3921.

[15] J. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8778–8787.

[16] Z. Huang, H.-X. Chen, T. Zhou, Y.-Z. Yang, C.-Y. Wang, and B.-Y. Liu, "Contrast-weighted dictionary learning based saliency detection for VHR optical remote sensing images," *Pattern Recognit.*, vol. 113, May 2021, Art. no. 107757.

[17] D. Zhao, J. Wang, J. Shi, and Z. Jiang, "Sparsity-guided saliency detection for remote sensing images," *J. Appl. Remote Sens.*, vol. 9, pp. 1–14, Sep. 2015.

[18] L. Zhang, S. Wang, and X. Li, "Salient region detection in remote sensing images based on color information content," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 1877–1880.

[19] L. Zhang, Y. Wang, and Y. Sun, "Salient target detection based on the combination of super-pixel and statistical saliency feature analysis for remote sensing images," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 2336–2340.

[20] L. Zhang, Y. Liu, and J. Zhang, "Saliency detection based on self-adaptive multiple feature fusion for remote sensing images," *Int. J. Remote Sens.*, vol. 40, no. 22, pp. 8270–8297, Nov. 2019.

[21] C. Li, R. Cong, J. Hou, S. Zhang, Y. Qian, and S. Kwong, "Nested network with two-stream pyramid for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9156–9166, Nov. 2019.

[22] Q. Zhang et al., "Dense attention fluid network for salient object detection in optical remote sensing images," *IEEE Trans. Image Process.*, vol. 30, pp. 1305–1317, 2021.

[23] C. Li et al., "A parallel down-up fusion network for salient object detection in optical remote sensing images," *Neurocomputing*, vol. 415, pp. 411–420, Nov. 2020.

[24] K. Shen, X. Zhou, B. Wan, R. Shi, and J. Zhang, "Fully squeezed multiscale inference network for fast and accurate saliency detection in optical remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] X. Zhou, K. Shen, Z. Liu, C. Gong, J. Zhang, and C. Yan, "Edge-aware multiscale feature integration network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5634819.

[26] Z. Huang, H. Chen, B. Liu, and Z. Wang, "Semantic-guided attention refinement network for salient object detection in optical remote sensing images," *Remote Sens.*, vol. 13, no. 11, p. 2163, May 2021.

[27] G. Li, Z. Liu, Z. Bai, W. Lin, and H. Ling, "Lightweight salient object detection in optical remote sensing images via feature correlation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601111.

[28] Z. Tu, C. Wang, C. Li, M. Fan, H. Zhao, and B. Luo, "ORSI salient object detection via multiscale joint region and boundary model," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607913.

[29] X. Zhou et al., "Edge-guided recurrent positioning network for salient object detection in optical remote sensing images," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 539–552, Jan. 2023.

[30] R. Cong, Y. Zhang, L. Fang, J. Li, Y. Zhao, and S. Kwong, "RRNet: Relational reasoning network with parallel multiscale attention for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5613311.

[31] Y. Zhou, F. Wang, J. Zhao, R. Yao, S. Chen, and H. Ma, "Spatial-temporal based multihead self-attention for remote sensing image change detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6615–6626, Oct. 2022.

[32] B. D. Brabandere, X. Jia, T. Tuytelaars, and L. Van Gool, "Dynamic filter networks," in *Proc. Adv. Neural Inform. Process. Syst.*, 2016, pp. 1–14.

[33] J. He, Z. Deng, and Y. Qiao, "Dynamic multi-scale filters for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3561–3571.

[34] Y. Pang, L. Zhang, X. Zhao, and H. Lu, "Hierarchical dynamic filtering network for RGB-D salient object detection," in *Proc. Eur. Conf. Comput. Vis.*, A. Vedaldi, H. Bischof, T. Brox, and J.-M. Frahm, Eds. Cham, Switzerland: Springer, 2020, pp. 235–252.

[35] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[36] F. Perazzi, P. Krähenbühl, Y. Pritch, and A. Hornung, "Saliency filters: Contrast based filtering for salient region detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2012, pp. 733–740.

[37] K.-Y. Chang, T.-L. Liu, H.-T. Chen, and S.-H. Lai, "Fusing generic objectness and visual saliency for salient object detection," in *Proc. Int. Conf. Comput. Vis.*, Nov. 2011, pp. 914–921.

[38] Z. Wang, D. Xiang, S. Hou, and F. Wu, "Background-driven salient object detection," *IEEE Trans. Multimedia*, vol. 19, no. 4, pp. 750–762, Apr. 2017.

[39] H. Jiang, J. Wang, Z. Yuan, Y. Wu, N. Zheng, and S. Li, "Salient object detection: A discriminative regional feature integration approach," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2013, pp. 2083–2090.

[40] Z. Wu, L. Su, and Q. Huang, "Stacked cross refinement network for edge-aware salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7263–7272.

[41] Z. Li et al., "Dense attentive feature enhancement for salient object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 12, pp. 8128–8141, Dec. 2022.

[42] M. Zhang et al., "Dynamic context-sensitive filtering network for video salient object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 1533–1543.

[43] Y. Nirkin, L. Wolf, and T. Hassner, "HyperSeg: Patch-wise hypernetwork for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4060–4069.

[44] B. Zhang, Y. liu, Z. Tian, and C. Shen, "Dynamic neural representational decoders for high-resolution semantic segmentation," in *Proc. Adv. Neural Inform. Process. Syst.*, 2021, pp. 17388–17399.

[45] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*, 2015, pp. 234–241.

[46] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[47] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.

[48] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[49] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inform. Process. Syst.*, vol. 30, 2017, pp. 1–15.

[50] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[51] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[52] F. Yu, V. Koltun, and T. Funkhouser, "Dilated residual networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 636–644.

[53] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4558–4567.

[54] D.-P. Fan, C. Gong, Y. Cao, B. Ren, M.-M. Cheng, and A. Borji, "Enhanced-alignment measure for binary foreground map evaluation," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 698–704.

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[55] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[56] Z. Deng et al., "R^3Net: Recurrent residual refinement network for saliency detection," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, Jul. 2018, pp. 684–690.

[57] X. Hu, L. Zhu, J. Qin, C.-W. Fu, and P.-A. Heng, "Recurrently aggregating deep features for salient object detection," in *Proc. AAAI*, 2018, pp. 6943–6950.

[58] M. Jing, D. Zhao, M. Zhou, Y. Gao, Z. Jiang, and Z. Shi, "Unsupervised oil tank detection by shape-guide saliency model," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 3, pp. 477–481, Mar. 2019.

[59] Q. Zhang, L. Zhang, W. Shi, and Y. Liu, "Airport extraction via complementary saliency analysis and saliency-oriented active contour model," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 7, pp. 1085–1089, Jul. 2018.

[60] G. Li, Z. Liu, W. Lin, and H. Ling, "Multi-content complementation network for salient object detection in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5614513.

[61] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, and P. Torr, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2021.

[62] Y.-H. Wu, Y. Liu, L. Zhang, M.-M. Cheng, and B. Ren, "EDN: Salient object detection via extremely-downsampled network," *IEEE Trans. Image Process.*, vol. 31, pp. 3125–3136, 2022.

[63] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[64] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4510–4520.

[65] X. Li et al., "Semantic flow for fast and accurate scene parsing," in *Proc. Eur. Conf. Comput. Vis.*, Aug. 2020, pp. 775–793.

[66] C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, and N. Sang, "BiSeNet v2: Bilateral network with guided aggregation for real-time semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3051–3068, Nov. 2021.

[67] S. Zheng et al., "Rethinking semantic segmentation from a Sequence-to-Sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6877–6886.

[68] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inform. Process. Syst.*, 2021, pp. 12077–12090.



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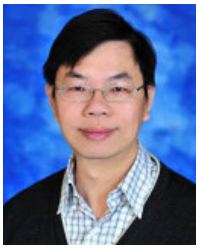
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